

The Beaufort Scale

This scale of wind speeds devised in 1805 by Sir Francis Beaufort is still in use today.

Force	Speed		Conditions
	knots	km/h	
0	<1	<1	Calm, sea like a mirror.
1	1-3	1-5	Light air, ripples only.
2	4-6	6-11	Light breeze, small wavelets (0.2m). Crests have a glassy appearance.
3	7-10	12-19	Gentle breeze, large wavelets (0.6m), crests begin to break.
4	11-16	20-29	Moderate breeze, small waves (1m), some white horses.
5	17-21	30-39	Fresh breeze, moderate waves (1.8m), many white horses.
6	22-27	40-50	Strong breeze, large waves (3m), probably some spray.
7	28-33	51-61	Near gale, mounting sea (4m) with foam blown in streaks downwind.
8	34-40	62-74	Gale, moderately high waves (5.5m), crests break into spindrift.
9	41-47	76-87	Strong gale, high waves (7m), dense foam, visibility affected.
10	48-55	88-102	Storm, very high waves (9m), heavy sea roll, visibility impaired. Surface generally white.
11	56-63	103-118	Violent storm, exceptionally high waves (11m), visibility poor.
12	64+	119	Hurricane, 14m waves, air filled with foam and spray, visibility bad.

Wave heights quoted are approximately those that may be expected in the open sea. In enclosed waters the waves will be smaller and steeper. Fetch, depth, swell, heavy rain and tide will also affect their height, and there will also usually be a time lag between any increase in the wind and the consequent increase in the sea.

to devise

to make up; to design

a breeze

a light wind

a calm

a quiet period

light

not heavy

ripple

a very small wave

gentle

mild

a wavelet

a not very big wave

moderate

average

a crest

the highest part of a wave

fresh

brisk

a white horse

a big wave with white crests

a hurricane

violent storm

appearance

the look of

violent

very strong

mounting sea

waves increasing

visibility

how far/well you can see

spray

small drops of liquid blown through the air

impair

to reduce

affect

to influence

foam

small white bubbles

bad

not good



Education and Culture

Leonardo da Vinci

a streak

thin line

poor

weak

downwind

same direction as wind

steep

goes up or down quickly 

upwind

against the wind

fetch

waves reflected back

spindrift

foam or spray blown from the surface of the sea

a depth

how deep the sea is

a swell

the heaving of the sea remaining after a storm

generally

usually; commonly

a time lag

the time between; delay

exceptionally

unusually

enclosed waters

surrounded by land

approximately

about

to quote

to mention



Education and Culture

Leonardo da Vinci